

NASA Turbofan Level B2: Data pretreatment and Modelling Plan

- codes: https://github.com/drvojtex/ADAML_NASATurbofanLevelB2

Data pretreatment

As a first step of data pretreatment, the constant columns (Sensors 1, 5, 6, 10, 16, 18 and 19) were removed. Next, the dataset Unit's curves were randomly shuffled and divided into the three groups. The first group are training-calibration curves (60 % curves of the dataset), the second group are validation curves (20 %) and the third group are testing curves (20 %).

Next, there was performed a zscore standardization. It was performed in a way that the mean value and standard deviation were computed on the training-calibration data and these values are used for standardization of all three groups. So the validation and testing curves are standardized according to the training-calibration curves. Standardization removes the influence of different sensor scales, and all variables contribute to further calculations (e.g. PCA) with the same weight. Histograms of all data points after zscore standardization are shown by Figure 1.

Missing values in the dataset were not explicitly addressed, because the dataset does not contain any such records (samples with unknown values of one or more sensors).

Since the time-series curves are of different lengths, the regression will be performed by the window-based approach. For that reason, the curves were, as a next pretreatment step, split into windows. Such multidimensional windows will be used as samples. To preserve context information, to each window (and subsequently to each variable, except the target variable) were attached the slope and intercept of the linear approximation of the previous window (within the curve). To the first window of each curve was attached zero context information. Based on this, each window (except the first one of the curve) is not processed as an isolated order set of points, but as an order set of points with information about the previous linear trend approximation. Such an example of a divided curve (data of Sensor 3) is shown by Figure 2. The particular samples (Sensor 3 values, intercept and slope) are distinguished by colours. The dashed line shows a context from the previous window, and the points show particular window sensor values.

If we set the stride used for obtaining windows to 1, the window size to 11 then each sample will have 13 variables and each variable will assigned 13 values (windows size plus intercept plus slope). The size of the data groups for such setup will be as follows: training-calibration group 11905 samples, validation group 3495 samples, 3781 samples.

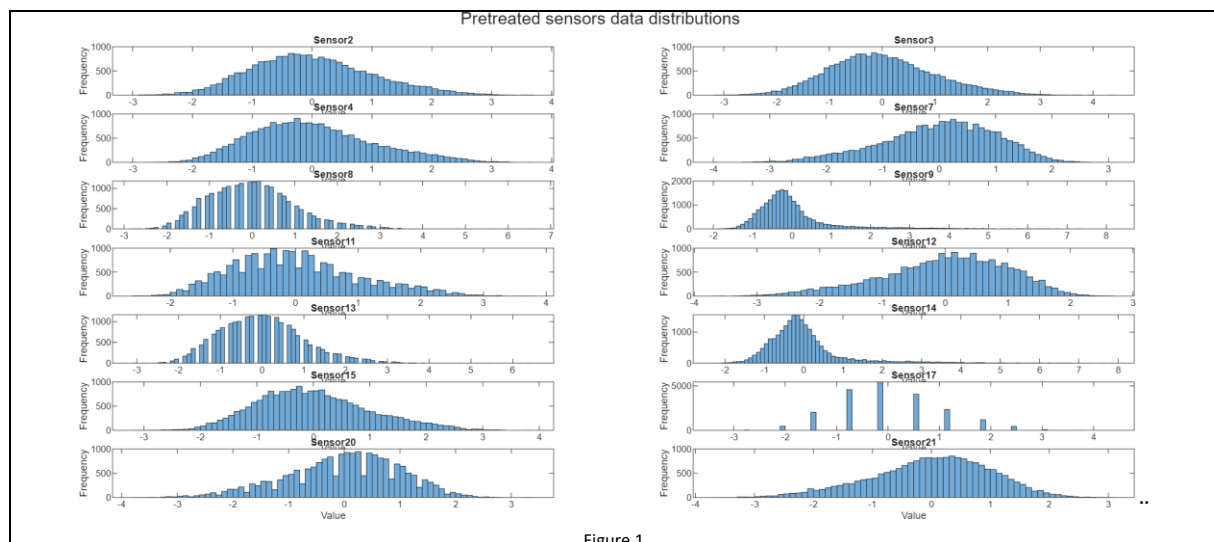


Figure 1

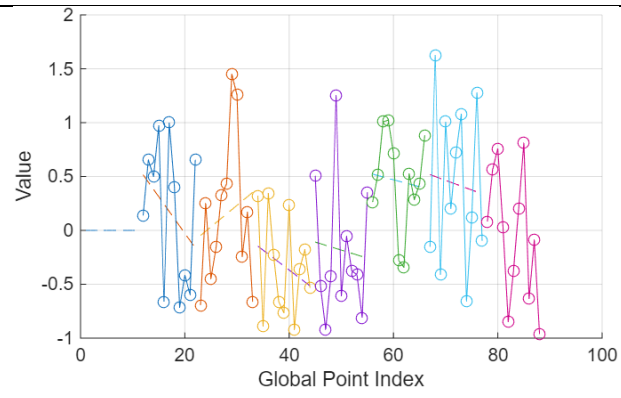


Figure 2

Modelling plan

Modelling goal

The aim is to perform multivariate regression (and evaluate its quality), where the input is a window of measured sensors values (part of the time-series), and the outputs are values of the Sensor 17 of the respective window. Furthermore, the task is also to identify the minimal subset of sensors and the optimal number of latent variables required to predict the target sensor.

Mathematical methods used in modelling

The regression will be performed on windows. Each window of length w will include sensor data and contextual information – intercept and slope of each sensor linear approximation of the previous window (within the curve). The windows will act as samples. Principal Component Analysis (PCA) will be applied to extract latent variables that capture the main sources of variability among the flattened input samples. Principal Component Regression (PCR [1]) will be performed using the most significant principal components to predict the target sensor values (based on the linear least squares). This approach not only reduces the dimensionality of the data but also mitigates multicollinearity among the sensor inputs. The key model hyperparameters are D (set of input sensors), window size w and k (number of PCs); which will be selected based on cross-validation performance. Mathematical formulation of setting PCR parameters is

$$\hat{\beta} = (\mathbf{P}_k^T \mathbf{X}^T \mathbf{X} \mathbf{P}_k)^{-1} \mathbf{P}_k^T \mathbf{X}^T \mathbf{Y},$$

where $\mathbf{P}_k \in \mathbb{R}^{[|D| \cdot (w+2)] \times k}$ is the orthogonal matrix of the first k principal components (each column is a principal component vector, which are eigenvectors of the covariance matrix of $\mathbf{X}$), $\mathbf{X} \in \mathbb{R}^{n \times [|D| \cdot (w+2)]}$ is the matrix of input features and $\mathbf{Y} \in \mathbb{R}^{n \times w}$ is the target matrix. Prediction for the new measurement $\mathbf{x}_{\text{new}}$ is given as $\mathbf{x}_{\text{new}}^T \mathbf{P}_k \hat{\beta}$.

Model calibration, validation and testing strategies

Model calibration, validation, and testing strategy will be performed using a combination of fixed splits and three-fold cross-validation. The input to the procedure will be clean data – zscore standardized window samples with addition of contextual information. The dataset will be randomly divided (while preserving particular curves – samples coming from the same curve will be in the same part) into five parts (1–5), with part 4 reserved as a fixed validation set (validation group) and part 5 as a fixed test set (testing group). Three-fold cross-validation is applied to parts 1–3 (training-calibration group) as follows: for each fold, two parts are used to train the PCR model for multiple combinations of hyperparameters (k , w and D), and one part is used as a calibration set to select the hyperparameter (k , w , D) combination with the best performance. After cross-validation, the fixed validation set (part 4) is used to select, among the three folds, the best-performing parameter combination (PCR parameters setup, k , w and D), and the final model is evaluated on the fixed test set (part 5) to provide an objective assessment of performance. Root mean squared error (RMSE) and R^2 will be used as the evaluation metrics. Figure 3 shows the strategy.

Next, after determining the optimal window size w and the set of sensors D , the effect of the number of latent variables (k) on R^2 and RMSE will be explored, to identify the dependence of the model complexity on its performance. Furthermore, the autocorrelation of the regression prediction residuals will be analysed, to verify that the residuals behave as white noise and that no systematic temporal patterns remain unmodeled. A comparison of selected time series (predictions versus ground truth) will also be conducted, to visualise the model's ability to reproduce the trend of the missing sensor values based on the observed data. Finally, a biplot will be generated to display the latent variables in relation to the samples, to interpret the latent space and reveal relationships between variables, samples, based on the regression model.

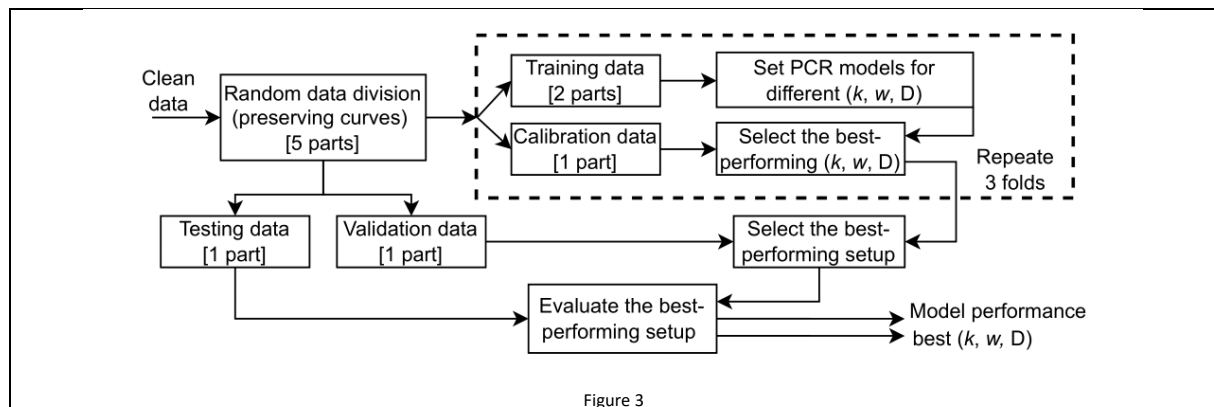


Figure 3

[1] DOI: 10.1002/0471725331